Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

MODEL LAB

Date: 16-04-2026

React JS Based Web Application for Internal Event Ticket Booking

SDG: SDG-4 QualityEducation.

Presented By:

T.VENKATA PAVANI– VTU25989 – 23UECS0563

Courses Handling Faculty:

Dr.Hafizur Rahman-Assistant Professor

INTRODUCTION

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This project focuses on automating the event registration process for department fests and seminars. It eliminates the need for manual ticket issuance and physical record-keeping.
The application provides a real-time interface for students and faculty to view event specifics.
Importance lies in reducing administrative overhead and preventing overbooking of venues.
The system ensures data accuracy through automated validation and state management. It offers a user-friendly platform accessible via any modern web browser

PROBLEM STATEMENT

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Manual event booking systems are prone to human error and data inconsistency.
Tracking "Available vs. Booked" tickets in real-time is difficult without automation.
Users often lack a central platform to view event details like venue, price, and timing.
Paper-based systems are inefficient and lead to delays in generating booking summaries. The application provides a real-time interface for students and faculty to view event specifics.

OBJECTIVE(S)

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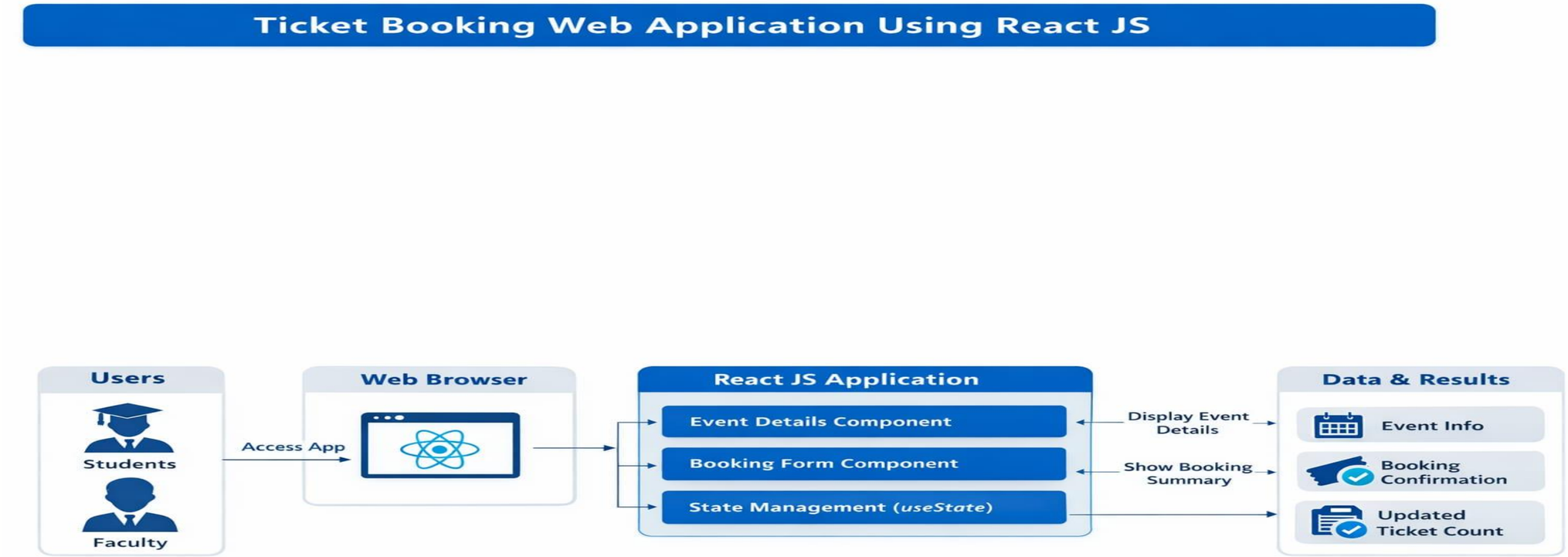
- To design and develop a responsive React JS web application for internal department event booking.
- To implement real-time ticket availability tracking and automated booking summaries.
- Ensure smooth user experience.

PROPOSED SYSTEM

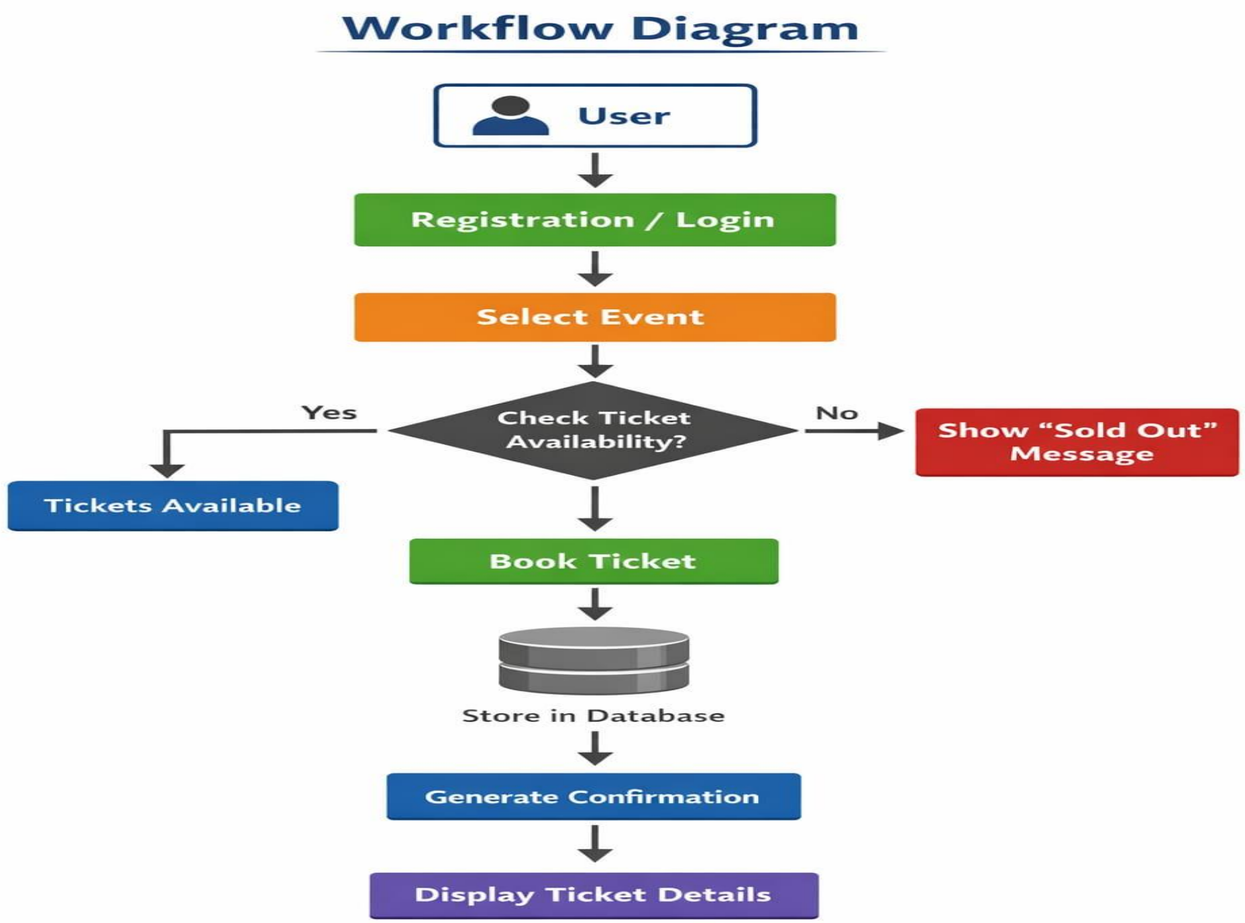
- Online ticket booking system for students and staff.
- User registration and login functionality.
- Real-time ticket availability tracking.
- Secure data storage using database.
- Automatic booking confirmation.
- Easy cancellation and updates.
- Reduces manual errors and saves time.

PROPOSED SYSTEM ARCHITECTURE



- **Input**: The user provides their name, email ID, department, and the specific number of tickets required for the event.
- **Output:** The system displays a real-time updated ticket count and a booking summary containing the user's name, event name, total tickets booked, and the final amount.

WORKFLOW


Workflow Diagram
User
Registration / Login
Select Event
Check Ticket Availability?
Yes
No
Tickets Available
Show "Sold Out" Message
Book Ticket
Store in Database
Generate Confirmation
Display Ticket Details

TECHNOLOGIES USED

- Frontend: React JS, JSX, CSS3.
- Tools: VS Code, Node.js (npm), Browser Developer Tools.
- Hardware/Software Requirements : Standard PC/Laptop with 4GB+ RAM.
- Hooks: useState for state management.
- Software: Windows/Linux/macOS, Modern Browser (Chrome/Firefox).

MODULE DESCRIPTION

Module 1: Event Details:
Displays static information like event name, venue, and price.
Module 2: Ticket Booking:
Captures user input and handles the "Book Ticket" event logic.
Module 3: Validation Engine: Validation Engine:
Ensures email format is correct and ticket count is positive.
Module 4: Summary & Analytics:
Calculates total cost and shows a success summary after booking

IMPLEMENTATION SCREENSHOTS

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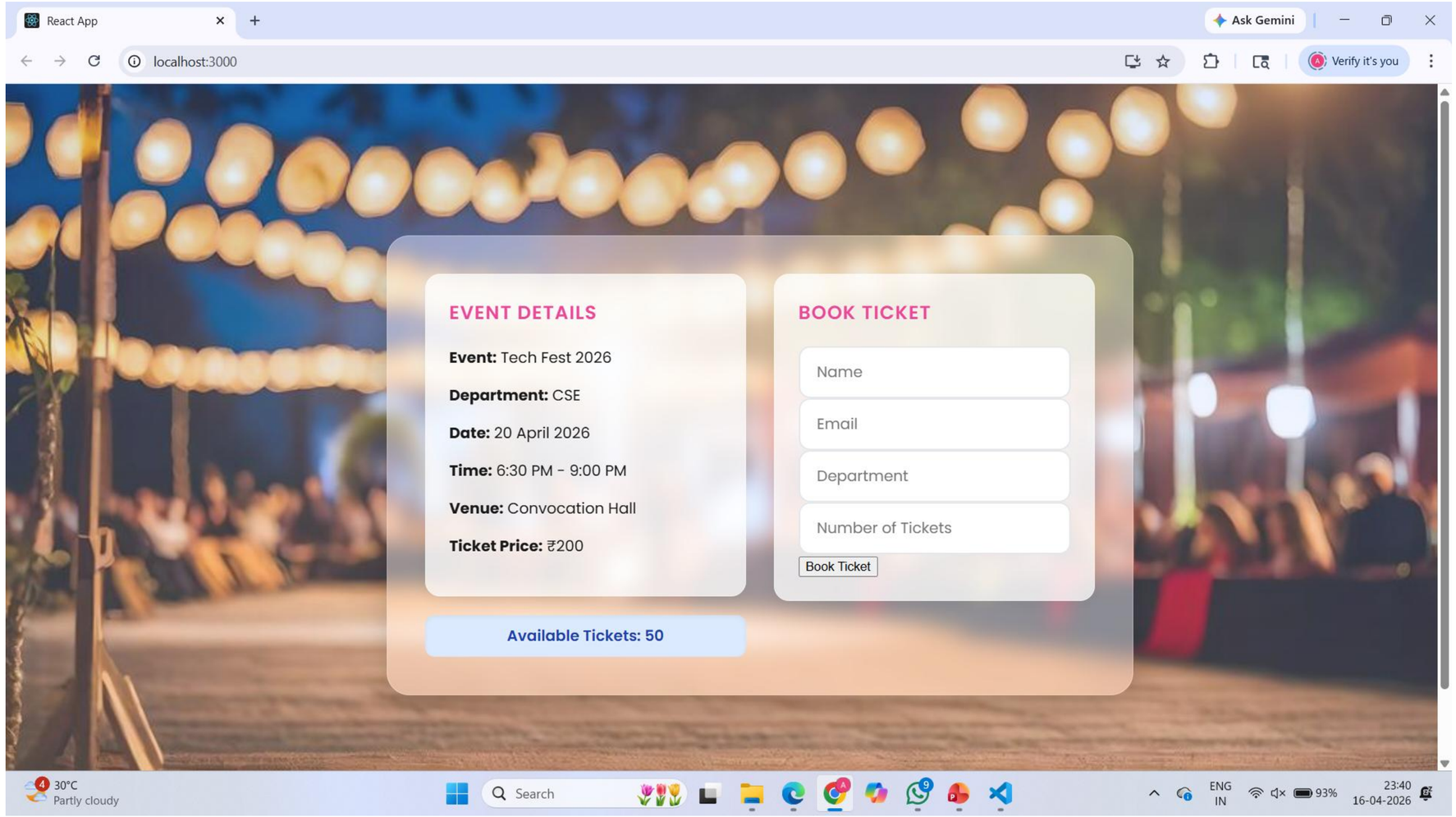

IMPLEMENTATION SCREENSHOTS

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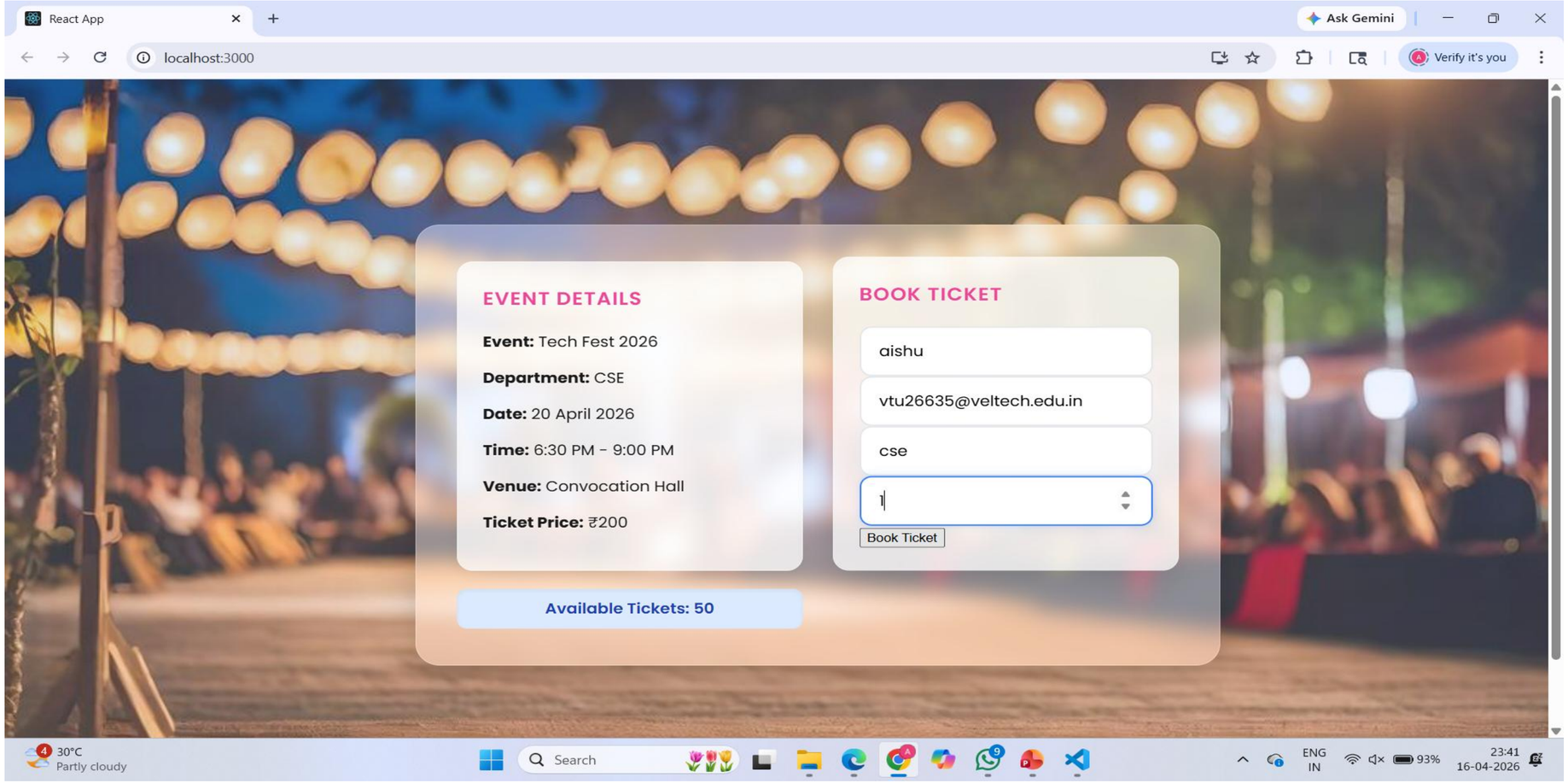

IMPLEMENTATION SCREENSHOTS

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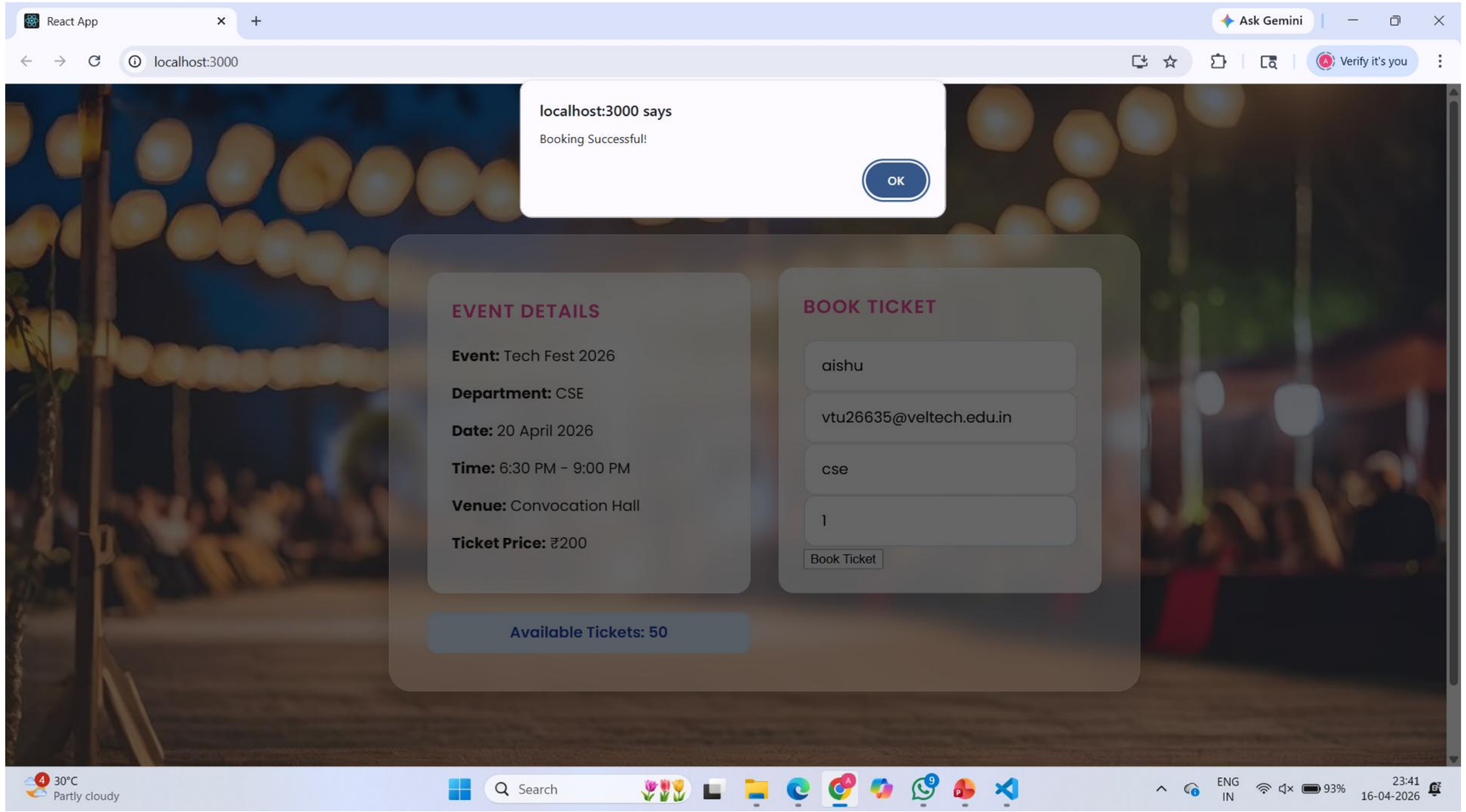

IMPLEMENTATION SCREENSHOTS

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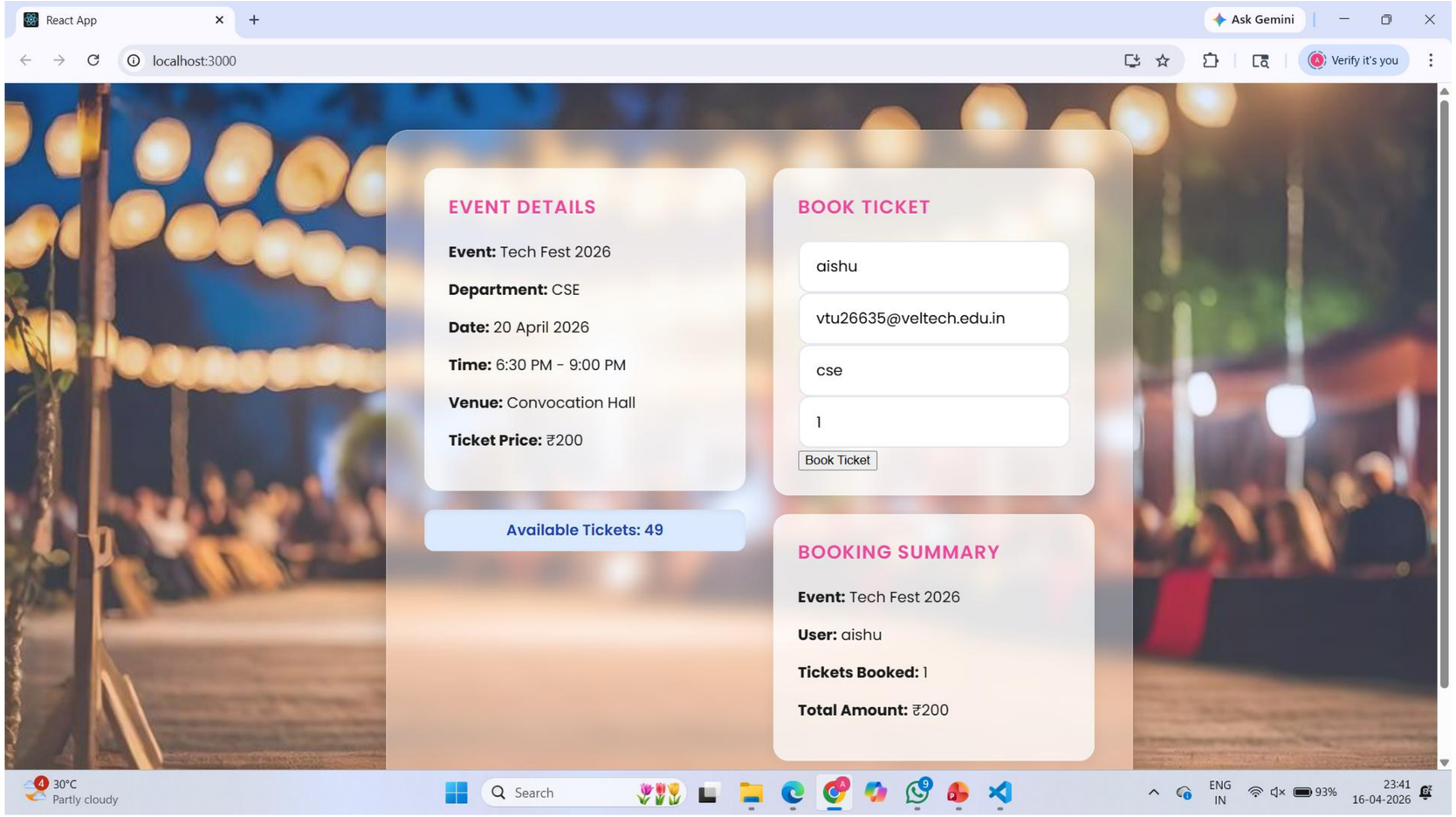

ADVANTAGES & LIMITATIONS

ADVANTAGES:

- Prevents overbooking through strict logic checks.
- Clean, modular code structure using functional components.

LIMITATIONS:

- Data is lost on page refresh (no persistent backend database yet).
- Limited to single-event booking in the current version

Thank you